EPR Generation Time and Dynamic Pre-generation: Study Briefing

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Status. The experiments are not finished. Results marked *interim* can change. The full report comes later.

1. The question

On-demand generation (ODG) creates an EPR pair when a request needs it, so the request waits for it. Dynamic is the compiler-driven strategy: it uses the known future requests to create the pair earlier, so the request can find it ready. **The idea under test:** if generating a pair takes longer, ODG suffers more, so Dynamic should look better. This study asks what the generation time really is, which parameters change it, and when Dynamic gains from it.

2. The baseline

Every experiment changes only one or two things from this baseline.

Parameter	Baseline	Meaning
Architecture	4×4 mesh, 16 cores	12 slots per core: 8 data and 4 communication
Memories per core, M	3 or 4	communication memories available for EPR pairs
Compiler memories, C	$M - 1$	at most this many prepared pairs per core; one slot stays for the ODG fallback
Memory efficiency	0.95	chance a memory emits its photon
Detector efficiency	0.95 each	chance each of the two detectors fires
BSM success rate	0.5	linear-optics limit
Fibre	0.025 m, 0.0002 dB/m	loss is negligible at this length
Success per attempt, p	0.407	$0.5 \times (0.95 \times 0.95)^2$
Classical delay, δ	100 µs per message	taken from a metro-scale paper, too slow for chip scale
Coherence time	1 s	how long a stored pair stays usable
Empty-layer gap	5 ms	time charged for a layer with no inter-core request; a leftover legacy constant, not physics
Request window, retry	50 ms, 1 ms	fixed absolute values

Table 1: Physical and timing baseline.

Compiler option	Baseline	Meaning
Lead	2 layers	the planner first tries to start a pair 2 layers before its request
Lookahead	8 layers	how far back it may look if that layer is full
Generation capacity	C	preparations that may start per core per layer
Compiler reservation	50 ms	how long a prepared pair may hold its memory slot
Chain hops	all	every hop of a multi-hop move is prepared
Admission	ready-first	requests with a ready pair go first
Consumption	handoff	the request takes over the prepared pair in place

Table 2: Dynamic baseline, the same as the 2026-10-30 routing study except $C = M - 1$.

Workload	Qubits	Layers	With traffic	Empty	Hops
QFT (main)	96	766	229	537	3833
QAOA (main)	96	733	328	405	7913
Draper adder	96	950	291	659	2661
CDKM adder	48	715	169	546	402
Random circuit	48	3341	2217	1124	10336

Table 3: Workloads. Mappings are frozen, so a route is the only difference between two traces of a workload.

Routes. XY is the base. `history` is the project's chosen optimal route: longest moves first, shortest paths only, choosing the path that adds least overload, with a small penalty for cores used in earlier layers.

Metrics. *Latency* is the mean per-hop time from eligible to EPR delivered. *Makespan* is the time to finish the workload, including the empty-layer gaps. *Communication time* is makespan with those gaps removed. *Hit* means the pair was ready before the request was first submitted. *Gain* is $100 \times (1 - \text{Dynamic/ODG})$, so a negative gain means Dynamic is worse. Data teleportation and local gates are not simulated.

3. What the generation time is

Result: mean generation time $= (4 \text{ or } 5) \times \delta \,/\, p$. This was measured on the real simulator, about 330,000 pairs over 108 probes.

Each attempt is a chain of classical messages. The diagram shows measured timelines for a forward hop, where the source core has the lower number.

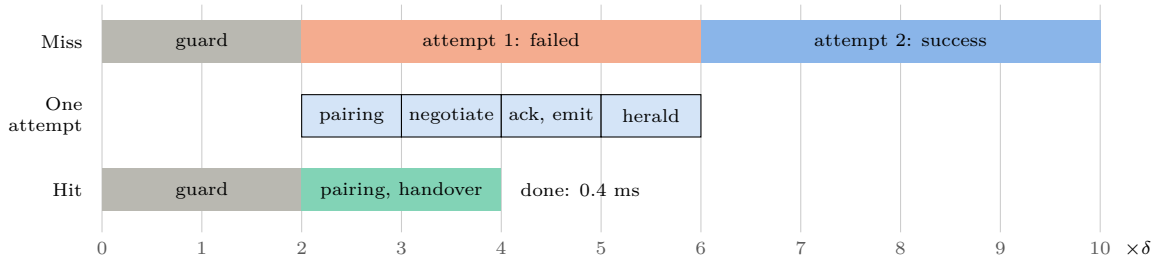


Figure 1: Guard is the reservation REQUEST and APPROVE. A failed attempt repeats everything, including pairing. A reverse hop adds one δ to each attempt and to a hit.

- **Hop direction matters.** An attempt takes exactly 4δ (0.4 ms) when the source core number is lower and 5δ (0.5 ms) when it is higher, because the node that sends NEGOTIATE then starts one message later.
- **Mean attempts** are $1/p \approx 2.46$. So generation takes 0.98 ms forward and 1.23 ms reverse, about 1.1 ms on average.
- **A hit is not free.** It costs 0.4 ms forward or 0.5 ms reverse, with no physical attempts.
- **Correction.** Earlier I said "0.4 ms per attempt". That holds for only one direction.

4. The knobs

Knob	Measured effect	Note
Classical delay δ	scales generation and the guard together	$\times 5$ gives 5.44 ms
Memory efficiency	changes attempts only	0.8 gives 1.56 ms, 0.4 gives 5.97 ms
Detector efficiency	same as memory efficiency	enters squared
BSM success rate	changes attempts only	0.2 gives 2.64 ms
Fibre length	changes both delay and loss	matters at km scale only: 10 km to the BSM gives 5.50 ms
Attenuation	changes attempts only	needs $\sim$100 dB/m at 2.5 cm
Memory frequency	none at 80 MHz	limits attempts only below about 2 kHz
Coherence, raw fidelity, gate and measurement fidelity, gap	no effect on generation time	coherence and raw fidelity change delivered fidelity; at coherence 0.2 ms it is 0.43

Table 4: Single-link results, mean of both hop directions.

5. Findings so far (interim)

(a) With the 5 ms gap, your idea holds. Latency gain rises as generation slows.

Case, `history`	Eff.	ODG (ms)	Dyn. (ms)	Latency	Makespan	Comm.	Hit rate
QFT $M4$ $C3$	1.00	1.211	0.675	44.2%	11.4%	40.9%	73.6%
	0.95	1.322	0.703	46.8%	13.0%	43.6%	73.6%
	0.80	1.776	0.819	53.9%	18.0%	48.3%	73.4%
QAOA $M4$ $C3$	0.95	1.364	0.981	28.0%	11.4%	22.7%	53.7%
QAOA $M3$ $C2$	0.95	1.454	1.265	13.0%	4.9%	9.7%	38.1%

Table 5: Ten seeds. At baseline, misses come from the planner's memory limit: almost every planned pair arrives in time.

(b) A pair waits far longer than it takes to make: about 1.1 ms to prepare, then about 9 ms in storage, because a two-layer lead usually spans two 5 ms gaps.

(c) The most important result so far: Dynamic's advantage depends on that 5 ms gap.

QFT $M3$ $C2$, `history`	Efficiency	Gap (ms)	ODG (ms)	Dynamic (ms)	Latency gain	Hit rate
Normal generation	0.95	5.0	1.357	0.933	+31.3%	55.2%
	0.95	0.5	1.357	1.472	−8.5%	27.4%
	0.95	0.0	1.357	1.780	−31.2%	16.1%
Slow generation	0.60	5.0	3.098	1.808	+41.6%	52.4%
	0.60	0.0	3.097	3.588	−15.8%	17.9%
Very slow generation	0.40	0.0	6.722	6.752	−0.5%	19.7%

Table 6: Four seeds, partial. QAOA $M3$ $C2$ at gap 0 is similar: −26.1% at efficiency 0.95.

This matters because 5 ms is not a physical value. With no gap, default Dynamic is worse than ODG, and a slower generation only reduces how much worse. So far this is with lead 2 and default options. **I do not yet know why it gets worse**, or whether better settings fix it. The lead and tuning experiments test that next.

6. What simulations are running

One *case* is one setting, for example: QFT, `history` route, $M = 4$, $C = 3$, memory efficiency 0.6. For each seed, it simulates the whole workload with ODG and again with Dynamic. Each of those is one *simulation*. Every hop is physically simulated, including every failed attempt, and an audit checks that no physical rule was broken.

#	Experiment	What it changes	What it answers	Done
1	gap (running)	empty-layer gap 0 to 20 ms	how much gain comes from the 5 ms constant	12/60
2	efficiency	memory efficiency 1.0 to 0.3, XY and `history`	gain against generation time	40/72
3	lead_gap	lead 1 to 8, gap 0 and 5 ms	does a different lead rescue Dynamic?	0/48
4	tuning	27 Dynamic options, one at a time	which knobs truly help	0/220
5	delay	classical delay $\times 0.1$ to $\times 5$	the same question through δ	8/40
6	memory	$M = 2$ to 8	how much memory is worth	1/20
7	coherence	coherence 1 s to 1 ms	fidelity and pair expiry limits	0/24
8	routes	XY, greedy, `history`, `minhop`	is `history` really best at every speed?	0/24
9	workloads	Draper, CDKM, random circuit	do results hold on other circuits?	0/12
10	chip	fast δ with short gaps	the chip-scale regime	0/16
11	scale	all time constants together	do only ratios matter?	0/12
12	equivalence	three causes of the same p	is p one axis?	0/6
13	validation	best tuned options on unseen seeds	guards against tuning to noise	later

Table 7: Remaining: about 3,100 simulations. Single-link screening is finished, except three reruns.

Ten hypotheses were committed before the sweeps ran, each with a way to fail. They are in the repository at `docs/research/generation-time-2026-11-01/plan.md`, with a logged list of every deviation.

7. Experiments we could add

Not running yet. These are options to discuss after the main results.

- **Realistic gap.** Replace the 5 ms constant with a time based on real local-gate durations.
- **Slack-aware planner.** Choose the lead from time available in ms, not a fixed layer count. This changes the planner.
- **No re-pairing per attempt.** Today a retry repeats pairing. Keeping the pairing would shorten attempts. This changes the protocol.
- **Fix the direction effect.** Let the node that starts first act as primary, so every attempt costs 4δ.
- **Enforce a fidelity threshold.** The trace uses 0.01, so no pair is ever rejected. A real threshold would test Dynamic's storage cost.
- **More mappings and circuit seeds,** because each main workload has only one mapping today.
- **Torus topology** and **data teleportation timing,** which are not modelled in this study.

8. Time estimate

With the Mac awake, plugged in and the other apps closed, cases now take about 2 to 4 minutes. The experiments should take about **2.5 to 3.5 hours**. Analysis, figures and the full LaTeX report then take about 1 to 1.5 hours more, so about **4 to 5 hours** in total. A closed lid pauses everything, and that time is added on top.